INTERNATIONAL MONETARY FUND

NIGER

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REQUEST FOR A THREE-YEAR ARRANGEMENT UNDER THE EXTENDED CREDIT FACILITY—DEBT SUSTAINABILITY ANALYSIS

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Prepared by the staffs of the International Monetary Fund and the International Development Association.

Niger: Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	*Moderate*
Overall risk of debt distress	*Moderate*
Granularity in the risk rating	*Limited space to absorb shocks*
Application of judgement	*No*

Niger's risk of external and overall public debt distress is assessed "moderate"—unchanged from the previous DSA.[1] [2] The response to the COVID-19 pandemic required higher borrowing and the shock entailed a sharp fall in exports, exacerbating the impact of Nigeria's decision to close its border to trade in 2019. Nevertheless, debt indicators remain below their thresholds thanks to concessional financing from donors and the prospective post-COVID rebound. Sustainability should be buttressed by the envisaged implementation of the government's reform program and the onset of crude oil exports via a new pipeline. In the medium and long run, further strengthening debt management, mitigating fiscal risks from SOEs and PPPs, prioritizing concessional borrowing, and strengthening private-sector development to support economic diversification would be key to strengthening Niger's debt sustainability.

[1] Niger's debt-carrying capacity remains rated "medium" with a composite indicator value of 2.96.

[2] Staff simulated a scenario assuming full disbursement of annual IDA allocations under credit terms, which did not affect the risk of external debt distress rating.

PUBLIC DEBT COVERAGE

1. The coverage of the public sector in the DSA is in line with the fiscal accounts and the previous DSA (Text Table 1). It covers the central government but excludes local governments and the social security fund. There are no extra budgetary funds. State guarantees extended to the private and public sectors for external borrowing are included. Publicly-guaranteed private debt is limited to the guarantee issued to the China National Petroleum Company (CNPC) for a loan to finance the government's minority stake in the refinery SORAZ.[3] SOEs do not directly borrow from abroad, benefitting instead from on-lending by the central government, which is captured in the debt statistics at the stage where the central government borrows the funds. This includes electricity (NIGELEC), water (SPEN), and telecom (Niger Telecom) companies, and the ABK, a public administrative entity set up for implementing the Kandadji dam project. Absent reliable data, the DSA cannot explicitly account for domestic SOE debt. The authorities are working with the World Bank to improve the availability and quality of financial information for SOEs. A dedicated directorate general has been established in the Ministry of Finance in late 2019. The authorities, under the World Bank's support, will publish certified financial statements for 2020 of at least nine out of 11 largest SOEs on the official website of the Ministry of finance to improve transparency and quality of debt reporting. External debt is defined on a currency basis.[4]

Text Table 1. Niger: Coverage of Public-Sector Debt and Design of Contingent Liability Stress Test

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	X
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	

	The country's coverage of public debt	The central government plus extra budgetary funds, central bank, government-guaranteed debt		
		Default	Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0 percent of GDP	0.0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	2.0	
4	PPP	35 percent of PPP stock	0.0	
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5.0	
	Total (2+3+4+5) (in percent of GDP)		7.0	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

2. The contingent liability tailored stress test is calibrated to account for debt coverage gaps (Text Table 1). First, the coverage shock is kept at 0 percent of GDP for other elements of the general government not captured in the baseline stock of debt since: (i) the authorities indicated that the strong financial position of the social security fund (CNSS) removes material fiscal risks; (ii) the authorities confirmed the absence of extra-budgetary funds; and (iii) local governments solely contract short-term debt with the domestic banking sector, which is small in size. Second, the contingent liabilities shock from SOE debt is set at the default value of 2 percent to reflect risks associated with their domestic borrowing

[3] CNPC extended a US$880 million (7.0 percent of GDP) loan for the construction of SORAZ refinery in 2008, of which US$352 million (2.8 percent of GDP) is guaranteed by the government. The outstanding stock of US$87.3 million (1.1 percent of GDP) at end-2020 is included in the baseline stock of debt and is expected to be fully repaid by 2023.

[4] The only exception is the borrowing from the West African Development Bank (BOAD), which is considered external debt despite being local-currency-denominated.

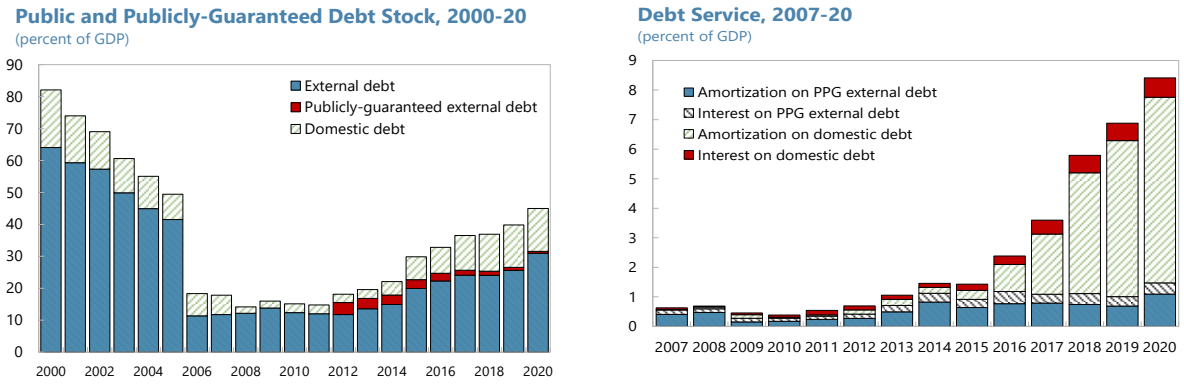
SOEs. Third, public-private partnerships (PPPs) signed under the new PPP law of May 2018 do not involve government financing—e.g., the recent renovation of the Niamey airport was fully privately financed in exchange for a 30-year concession to operate it. Projects under the previous PPP law are akin to installment payments and are reflected to the extent that they are implemented but yet paid. A contingent liability stress test for PPPs is hence not indicated at this time, and, taking into account credit to the economy of just 13.2 percent of GDP as of 2020, the default value of 5 percent of GDP for contingent liability risk appears adequate.

BACKGROUND

A. Evolution and Composition of Debt

3. Niger's public and publicly guaranteed (PPG) debt stood at 45.0 percent of GDP at end-2020 (Text Figure 1). External public debt dropped from an average of 52.7 percent of GDP during 2000–05 to 11.3 percent of GDP in 2006 following the debt relief granted under the HIPC and Multilateral Debt Relief Initiatives. It has risen slowly since 2012 as Niger developed natural resource projects and other large-scale public investments. Domestic debt had also been on a declining trend since 2000 and averaged 2.8 percent of GDP over 2008–14, before starting to rise from 2015. Debt service follows a similar pattern, with a large share of the amortization on domestic debt reflecting its shorter maturity. In 2020, the outstanding PPG debt increased at a higher pace than recent years as the heightened fiscal deficit in the context of the COVID-19 pandemic was financed mainly by external donor support.

Text Figure 1. Niger: Public and Publicly-Guaranteed Debt, 2000–20



Sources: Nigerien authorities, and IMF staff estimates.
Note: Domestic debt includes debt associated with commercial PPPs from 2017. In 2020 they stood at 0.6 percent of GDP.

4. PPG external debt makes up 70 percent of Niger's total debt stock in 2020. Multilateral creditors represent the lion's share (around four fifths) of external debt, with Niger borrowing most from the World Bank (IDA) followed by the BOAD and African Development Bank (AfDF). The official bilateral debt represents around one fifth of external debt. External debt exposed to exchange rate risk is relatively low (around one third of external debt) given the CFAF's peg to the euro. External debt is generally on concessional terms, with an average weighted interest rate of 1.2 percent and remaining maturity of 23 years at end-2020.

5. Domestic debt consists mostly of short- and medium-term Treasury securities, predominantly held by banks domiciled in Niger or in the rest of the West African Economic and Monetary Union (WAEMU). Outstanding Treasury securities remained almost flat in 2020 despite larger financing needs amid the pandemic as, the government carried out a debt reprofiling operation in January 2020, in which it borrowed CFAF 148 billion (1.9 percent of GDP) commercially from abroad to repay domestic debt.[5] In 2019 Niger placed government paper worth 3.9 percent of GDP for the first time through syndication. At seven years, the tenor was much above what is achievable in regional auctions. The average remaining maturity of Niger's domestic debt is 5.4 years while the average weighted interest rate stood at 4.1 percent at end-2020.

6. The estimation and analysis of private external debt is complicated by data issues and requires further follow-up. The regional central bank (BCEAO) does not yet compile private external debt stock statistics. Efforts to gather information on the coverage and composition of private external debt will continue, with technical support from the IMF's Statistics Department.

B. Macroeconomic Forecast

7. The baseline scenario is predicated on macroeconomic assumptions reflecting recent economic developments and the response to the COVID-19 pandemic, as well as ongoing and new policy measures (Text Table 2). While the direct health impacts of the pandemic have been limited, containment measures including the closure of the border with Nigeria slowed down economic activity in 2020, leading to a drop in exports and a sizeable increase in external financial assistance. With the removal of the remaining social distancing measures, the opening of the border with Nigeria, and the resumption of all infrastructure projects, growth is projected to pick up in 2021 and onward. Growth and exports are expected to be boosted in 2023 by the expected start of oil exports—one year later than in the previous DSA update due to the delay of the construction project amid the COVID-19 pandemic. Growth in the long run is projected at 6.1 percent in line with the previous DSA. Addressing constraints in the business environment and financial services, through the improvements to tax policy and administration and new mechanisms to support financial intermediation, is critical to enhancing private sector developments and reducing informality, thereby supporting stronger and more resilient growth. Ongoing reform on the education system is expected to improve human capital and contribute to the long-run growth. The exchange rate peg keeps inflation well-contained throughout the projection period.

8. The framework assumes gradual fiscal consolidation in line with the government's ongoing reform program. Given rising insecurity due to terrorism and crime across the Sahel, an increase in security-related spending is inevitable. Nonetheless, in line with the macro projections, a sizable bounce-back of revenues is expected for next years as the health crisis abates, the border with Nigeria is reopened to trade, and revenue measures regain traction, further strengthen by oil-related revenue when the exports commerce. Apart from the oil sector, the economy should also benefit from several other large projects related to stepped-up engagement of donors and foreign investors. Compared to the previous DSA, real GDP growth and the revenue-to-GDP ratio are

[5] The loan was contracted in January 2020 with Deutsche Bank for a total amount of euro 225 million, of which euro 179 million (1.5 percent of GDP) have been drawn. It was used to repay a set of five Treasury bills and one Treasury bond.

expected to be moderately lower in the coming years, reflecting persistently slow economic activities in formal non-agricultural sectors in the context of the pandemic and postponement of the onset of the oil exports from 2022 to 2023 due to the construction delay. The medium- and long-term projections remain almost unchanged.

Text Table 2. Niger: Key Macroeconomic Assumptions, 2018– 41

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027-41
Real GDP growth (percent)										
DSA 2021	7.2	5.9	3.6	5.4	6.5	10.4	11.4	8.5	6.0	6.1
Previous DSA	7.2	5.9	1.2	6.9	12.8	11.1	6.7	6.3	5.8	6.2
Inflation (CPI)										
DSA 2021	2.8	-2.5	2.9	2.9	2.5	2.0	2.0	2.0	2.0	2.0
Previous DSA	2.8	-2.5	2.8	0.4	2.0	2.0	2.0	2.0	2.0	2.0
Primary fiscal balance (percent of GDP)										
DSA 2021	-2.1	-2.6	-3.4	-3.6	-2.0	-1.3	-1.3	-1.3	-1.3	-1.1
Previous DSA	-2.1	-2.6	-3.4	-3.6	-1.9	-1.3	-1.3	-1.3	-1.3	-1.1
Total revenue excluding grants (percent of GDP)										
DSA 2021	12.1	11.2	10.8	10.9	11.5	12.7	13.4	13.8	14.0	15.5
Previous DSA	12.1	11.2	10.4	11.9	12.6	13.6	13.9	14.2	14.3	15.4
Exports of goods and services (percent of GDP)										
DSA 2021	11.3	10.8	16.6	14.5	14.8	17.3	19.5	20.8	19.6	18.8
Previous DSA	11.3	11.1	9.3	10.1	14.0	16.6	16.6	16.7	16.1	11.4
Oil export price (US dollars per barrel)										
DSA 2021	64.9	58.3	39.2	62.4	61.3	58.2	56.1	54.5	53.5	62.9
Previous DSA	64.9	58.3	39.6	44.4	45.7	46.7	47.7	48.6	49.6	53.9
Uranium price (Thousands of CFAF per kg)										
DSA 2021	40.8	44.0	48.7	46.9	46.2	46.1	46.0	46.0	46.0	46.0
Previous DSA	40.8	44.0	48.7	46.9	46.4	46.2	46.1	46.1	46.1	43.0

Source: IMF staff calculations.

9. Compared to the previous DSA, gold exports projections have been significantly revised upwards. Artisanal gold production has been booming in the Burkina Faso-Mali-Niger triangle in the last few years with Niger's gold industry benefitting from refining, trading, and shipment to abroad of gold imported from neighboring countries. The previous DSA update did not take this into account in the baseline as the BCEAO's data compilation process of Niger's gold exports in the balance-of-payments was ongoing—instead, an alternative scenario was assessed based on available customs data. The IMF provided a technical assistance (TA) through its statistics department to support the authorities' efforts to improve the quality of gold exports statistics. The data compilation has proceeded and is included in the baseline framework entailing an upward revision of exports and their projections[6].

10. The authorities aim to maintain limited reliance on domestic financing for years to come, with a view to extending maturities, reducing roll-over risk, and creating space for banks to lend to the private sector. Concessional and semi-concessional financing from external donors remain the main

[6] In accordance with the TA recommendation, gold exports by Nigerien refineries—accounting for the bulk of the increase in 2020 gold exports—are recorded as service exports as the country provides refining services almost exclusively to non-residents of neighboring countries.

sources of financing.[7] The share of domestic sources in total budgetary financing was lower than 5 percent in 2019 and 2020 thanks to the ample external financing in the context of large-scale project related loans and COVID pandemic as well as the debt reprofiling operation in January 2020. Over the medium term, it is projected to remain below 40 percent, slightly below the average in 2013-18 (42 percent). In the very long run, highly concessional donor support is assumed to gradually decline as a percent of GDP as the domestic financial market deepens.

11. The terms of foreign and domestic borrowing are assumed to shift over time to lower concessionality and longer maturities. For foreign debt, new disbursements are expected to be covered by external funding sources based on historical financing patterns. In the longer run, the weights of external creditors are adjusted so that external borrowing moves very gradually toward less concessional financing and toward commercial loans. For domestic borrowing, debt instruments are assumed to gradually shift from T-bills to medium- and long-term bonds. Consistent with the borrowing terms of recent government securities issuances on the regional market, the average interest rate on government bonds is assumed at 5.5, 6, and 6.5 percent for bonds maturing in 1 to 3, 4 to 7 years and over 7 years, respectively. The interest rate on T-bills is set to 5 percent.

12. The SDR allocation has given Niger access to more beneficial financing. The SDR allocation was on-lent in CFAF to member states by the BCEAO in the form of a 20-year loan carrying a fixed interest rate of 0.05 percent. The financing terms are more concessional than the arrangements for the 2009 SDR allocation (10-year repayment period with a three-year grace period and an interest rate of 3 percent). The arrangement is treated as domestic debt and evaluated in nominal terms in the DSA. This on-lending is expected to improve the overall debt profile by replacing more expensive domestic borrowing.

13. The DSA's tool kit to assess the realism of the macroeconomic forecast does not raise red flags in light of historical experience and comparison with peers.

a. **Drivers of debt dynamics** (Figure 3). The evolution of total public debt is dominated by developments of the primary fiscal deficit and real GDP growth. Compared to the past five years, projections put the contribution of growth higher and the one for the primary deficit lower. The rise of the public debt ratio is consequently arrested, and it stabilizes at around 40 percent of GDP, slightly higher than in the previous DSA. External public debt is projected to peak at 32.7 percent of GDP in 2021, and decline to 16.5 percent in 2041, entailing a similar pattern to total public debt. Unexpected changes in residuals were chiefly responsible for past forecast errors for total public debt whereas those in the primary deficit and the current account were for external public debt. Their magnitude is comparable to those in other low-income countries (LICs).

b. **Realism of planned fiscal adjustment** (Figure 4). The projected three-year fiscal adjustment in the primary balance (1.3 percentage point of GDP) lies slightly below the top quartile of the distribution of past adjustments to the primary fiscal deficit (above 2 percentage points of GDP) for a sample of LICs. The realism of the expected adjustment is predicated on the recovery from the pandemic (i.e., including one-off gain from the removal of COVID-related containment measures and constraints), the

[7] Concessional financing is defined as the one with grant element (GE) exceeding 35 percent of the face value.

improvements in non-oil revenue mobilization through expanding tax base and reducing tax expenditure, and the revenue boost from the start of crude-oil exports in 2023.

c. **Consistency between fiscal adjustment and growth** (Figure 4). The projected growth path for 2021 to 2022 is driven by the recovery from the pandemic, reinforced by the removal of the remaining restrictions, the full opening of the border with Nigeria, and the resumption of all infrastructure projects, with the impact of fiscal adjustments ranging from -0.5 to 0.5 percent of GDP based on different fiscal multipliers. With much of the fiscal adjustment being driven by higher oil-related revenues, the impact of consolidation on growth is likely to be muted.

d. **Consistency between public investment and growth** (Figure 4). The tool shows a similar share of public investment in GDP in the previous and the current DSAs. Private investment is expected to scale up in the projected years.

C. Country Classification and Determination of Stress Test Scenarios

14. Niger's debt-carrying capacity remains rated "medium". The methodology relies on a composite indicator (CI) combining the CPIA score,[8] external conditions as captured by global growth, and country-specific factors. Based on data from the April 2021 WEO vintage, the calculations give a CI value of 2.96, reflecting positive contributions from the CPIA (44 percent) but also international reserves (34 percent), and country and global real growth rates (6 and 14 percent, respectively) (Text Table 3). This score falls within the medium debt-carrying capacity thresholds defined as 2.69 < CI ≤ 3.05.

Text Table 3. Niger: CI Score

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	3.402	1.31	44%
Real growth rate (in percent)	2.719	6.983	0.19	6%
Import coverage of reserves (in percent)	4.052	45.974	1.86	63%
Import coverage of reserves^2 (in percent)	-3.990	21.136	-0.84	-29%
Remittances (in percent)	2.022	0.813	0.02	1%
World economic growth (in percent)	13.520	3.137	0.42	14%
CI Score			2.96	100%
CI rating			Medium	

Source : IMF staff calculations. The CI cutoff for medium debt-carrying capacity is 2.69 < CI ≤ 3.05.

15. The debt sustainability analysis relies on six standardized stress tests and a tailored commodity price shock stress test. The standardized stress tests use the default settings. While Niger does not qualify for the market financing shock stress test, the commodity price shock stress test is relevant and helps assess the sensitivity of projected debt burden indicators to unfavorable developments in commodity export prices.[9]

[8] The CPIA score is 3.42 in 2020.

[9] Under the debt sustainability framework, countries with commodity exports accounting for at least 50 percent of total exports of goods and services over the previous three-year period are subject to the stress test. Commodities accounted for 75.0 percent of Niger exports of goods and services over the period 2018-20.

DEBT SUSTAINABILITY

A. External Debt Sustainability

16. External debt is projected to fall gradually, with public and private debt both declining in the long run (Table 1). Under the baseline scenario, PPG external debt-to-GDP is expected to remain flat in 2021 owing to significant foreign borrowing to finance Niger's economic and social development agenda, followed by a downward trajectory to settle at 16.5 percent of GDP at the end of the projection period in 2041 as higher growth materializes and fiscal adjustments proceed. Total external debt displays a similar pattern—steadily declining to 24.0 percent of GDP in 2041. The non-interest current account deficit remains the main driver of these dynamics. The goods and services balance is projected to deteriorate until 2022 reflecting high imports related to the large projects, such as the Kandadji dam, a cement factory, a uranium mine, the oil export pipeline, and MCC-funded investments in agriculture. Coupled with growth in gold exports, production from these projects is expected to improve the current account sharply when they come on stream.[10] Once the non-interest current account deficit, net FDI, and endogenous debt dynamics are accounted for, remaining drivers of external debt dynamics, such as components of the capital account, reserves accumulation, valuation adjustments, as well as price and exchange rate changes, are subsumed into the residual.

17. The PPG external debt indicators remain below their thresholds throughout the projection period under the baseline scenario (Figure 1). The present value (PV) of debt-to-GDP is projected to follow a gradual downward trend over the projection period. The PV to exports ratio remains slightly below the threshold in 2021 and 2022 despite the severe impact of the COVID-19 pandemic on exports while it is projected to decline as exports strengthen due to the reopening of borders and the prospective oil exports via a new pipeline. Debt service to official creditors is low in 2021 benefiting from the G20 Debt Service Suspension Initiative (DSSI) and the IMF's Catastrophe Containment and Relief Trust (CCRT), which amount to 0.4 percent of GDP in 2020-21 and 0.3 percent of GDP through 2020-22, respectively. Both debt service to exports and revenue ratios are expected to enter a downward trajectory in the medium- and long-run after spikes in 2022.

18. Stress tests indicate that two export-related indicators (the PV of PPG external debt to exports ratio and debt service to exports ratio) exceed their threshold with export shocks and one-time depreciation shock temporarily bringing the debt service-to-revenue ratio above its threshold (Figure 1). Regarding the risk surrounding exports, while the oil pipeline project is expected to be completed by 2023, further delays cannot be ruled out. The exchange rate risk will remain contained in a currency union and given the CFAF's peg to the euro. Also, the PV of debt-to-GDP and exports ratios pick up later in the projected period under the historical scenario. The scenario mechanically captures large shocks such as draughts and floods, the evolution of macroeconomic variables with structural changes in the last ten years, large financing needs during the initial stages of extractive industries' projects and the political instability after the 2011 military coup, exaggerating the severity of the scenario.

[10] Around half of the increase in exports by 2026 is explained by the oil export pipeline.

19. The granularity assessment suggests that the space to absorb shocks is limited in 2021-22—unchanged from the previous DSA update—but space relative to the thresholds is expected to increase in the medium- and long-run (Figure 5). Under tests to qualify the moderate risk of external debt distress, the PV of debt-to-exports ratio currently falls into the limited space in 2021 and 2022, and the debt service-to-exports and revenue ratios enter the limited space in 2022, while they fall reaching to the territory of substantial space as exports pick up and revenue mobilization materializes in the next couple of years.

B. Total Public Debt Sustainability

20. Public sector debt is projected to decline gradually in the medium- and long-run (Table 2). After an increase during and after the pandemic to 49.8 percent of GDP in 2022, public debt is projected to decline to and stabilize at around 41.3 percent of GDP in the long run. This owes to growth recovery and fiscal consolidation after the pandemic, aided by higher oil revenues. In the medium and long run, improved domestic revenue mobilization, higher spending efficiency, and better expenditure control are expected to contribute to reducing the primary deficit. Gradual shifts toward lower concessionality and longer maturities over the medium- and long-run would marginally increase interest costs, while the benefit of the extension of maturities lies in reducing roll over risks.

21. The PV of the public debt-to-GDP ratio is below the benchmark in the baseline, though not in the event of an adverse commodity price shock (Figure 2). The PV of public debt-to-GDP ratio is well below the benchmark of 55 percent of GDP in 2021 and projected to gradually decline over the projection period under the baseline scenario. A commodity price shock would set the PV of debt and debt-service on a diverging path in the absence of compensating fiscal and other policy measures.[11]

C. Risk Rating and Vulnerabilities

22. Niger's risk of external and overall debt is rated "moderate" and debt is deemed sustainable. The moderate debt distress rating arises from the fact that no indicator for PPG external or public debt breaches its threshold under the baseline scenario. Debt remains sustainable as: (i) debt indicators remain on steady trajectories and overall public debt sustainability remains solid even in the face of adverse shocks; (ii) Niger's economy is expected to rebound from the pandemic, supported by the strengthening of the export base due to the onset of crude oil exports in 2023 and implementation of revenue mobilization measures in the medium run; (iii) the liquidity risk facing the country remains low since, as a member of the WAEMU, Niger can draw on the currency union's pooled external reserves, delinking the ability to service foreign debt from exports at the national level; and (iv) Niger is expected to continue to benefit from significant financial assistance from donors over the next few years to address the daunting development and security challenges it faces. In line with the Fund Debt Limits Policy (DLP), a debt limit on the new PPG external borrowing is embedded in program conditionality, and is calibrated to build an adequate buffer to avoid a downgrade of the risk of debt distress.

[11] The mechanics of the DSA could exaggerate the adverse effect of a commodity price shock. They set the growth of primary expenditure relative to historical averages of GDP growth for the years 2022 and 2023, thereby neglecting the high growth during the period in the baseline.

23. The authorities' sustained commitment to sound macroeconomic policies and economic reforms, as well as further progress on debt management in line with recent TA, will be critical to buttress debt sustainability. Progress in the following areas will be key:

- **Domestic revenue mobilization**. Niger's public debt relative to domestic revenues compares unfavorably to other WAEMU countries or Sub-Saharan Africa. Non-oil revenue mobilization is of fundamental importance with reducing tax expenditures and increasing the tax base taking a central position. Additional oil revenues expected in coming years should not be fully spent, thereby contributing to fiscal consolidation.

- **Fiscal risks and spending quality**. Possible mismanagement of large investments by SOEs or under PPPs have the potential to eventually add substantially to government debt. While Niger is in dire need to build up its infrastructure, proper evaluation and attention to good governance should not fall by the wayside in the rush. More generally, the government should try to make the most of limited resources by raising the efficiency of public spending, which remains low in Niger.

- **Economic diversification**. A narrow economic base and a low level of economic development generally are at the root of difficulties with mobilizing revenues and securing strong sustainable growth. Developing the local private sector including SMEs while tackling informality is paramount.

- **External borrowing**. Until export prospects are more certain, Niger should continue to prioritize external financing in the form of concessional loans and grants. Favoring euro-denominated debt, given the CFAF's peg to the currency, can also help reduce the exchange rate risk.

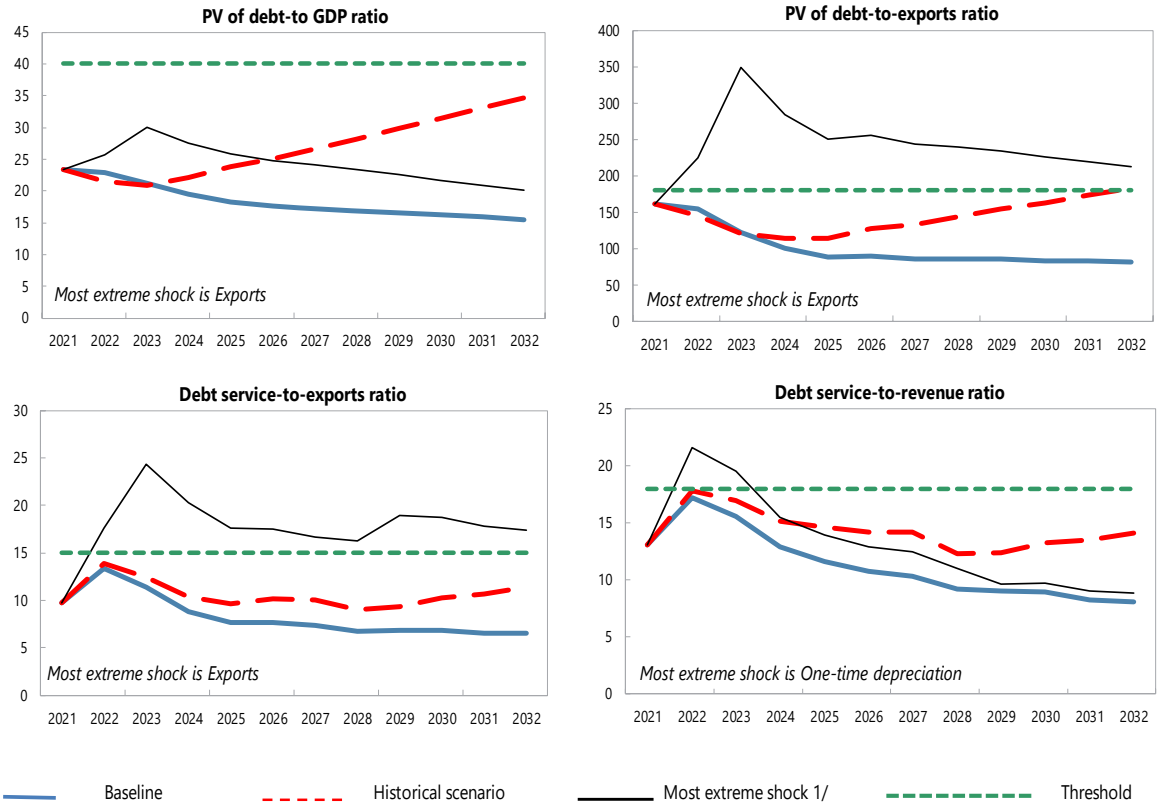
- **Domestic borrowing.** While benefiting from the current favorable market conditions supported by the accommodative regional monetary policy, dedicated market communication and marketing will be needed, in the medium- and long-run, to increase the average tenor of issuances and thereby reduce the vulnerability regarding refinancing. These efforts should be complemented by the implementation of a structural program of swaps of securities close to their maturity with securities of longer maturities.

- **Financing plan**. To avoid liquidity shortfalls and minimize the associated financing costs, not only the overall volumes of debt to be issued but also the issuance schedule for the year, along with effective instruments should be formulated in an annual financing plan. Integration with the annual cash flow plan would be essential.

D. Authorities' Views

24. The authorities agreed the conclusions of the DSA. They were pleased that Niger's moderate **rating** for debt distress is maintained thanks to concessional financing from donors and the prospective post-COVID rebound. They also agreed that it is essential to build an adequate buffer given the limited space to absorb shocks in the coming years and, in the medium- and long-run, further strengthening debt management, as well as sustained commitment to sound macro. policies and reforms will be critical.

Figure 1. Niger: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2021–32



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	No	
Natural Disasters	n.a.	n.a.
Commodity Prices 2/	No	No
Market Financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.7%	1.7%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	26	26
Avg. grace period	5	5

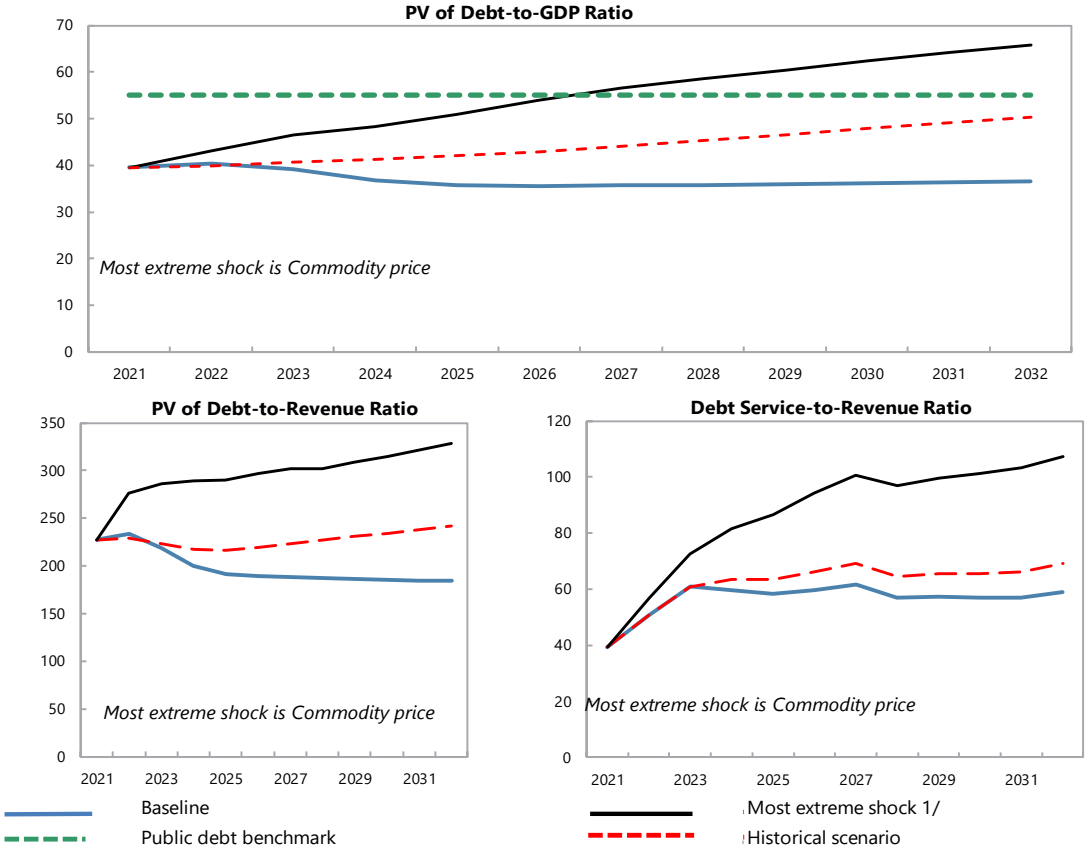
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Niger: Indicators of Public Debt Under Alternative Scenarios, 2021–32



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	22%	22%
Domestic medium and long-term	32%	32%
Domestic short-term	47%	47%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.7%	1.7%
Avg. maturity (incl. grace period)	26	26
Avg. grace period	5	5
Domestic MLT debt		
Avg. real interest rate on new borrowing	4.1%	4.1%
Avg. maturity (incl. grace period)	3	3
Avg. grace period	2	2
Domestic short-term debt		
Avg. real interest rate	3%	3.0%

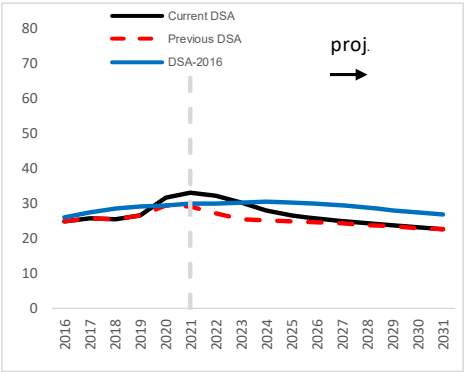
* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

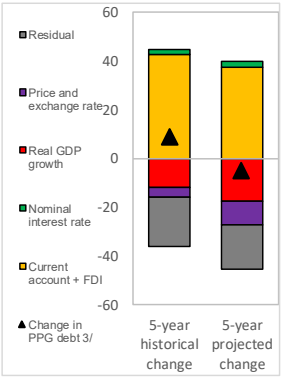
1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Figure 3. Niger: Drivers of Debt Dynamics—Baseline Scenario

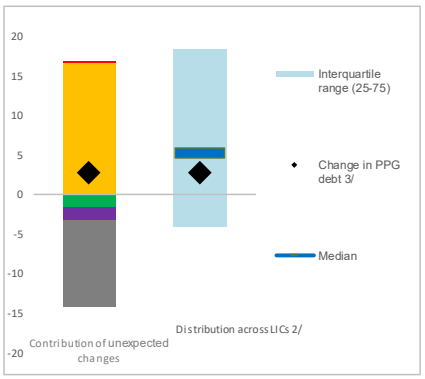
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

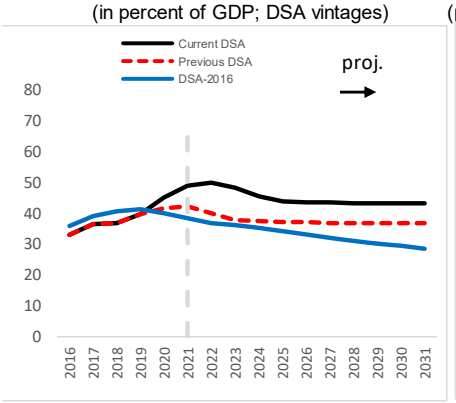


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

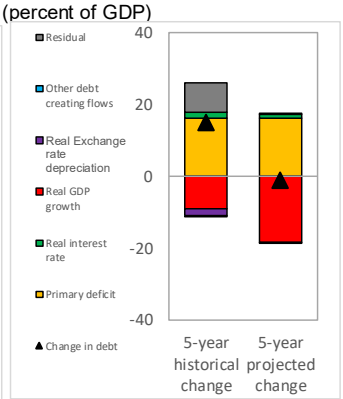


Public Debt

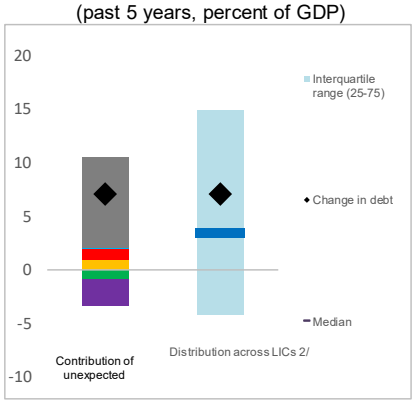
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



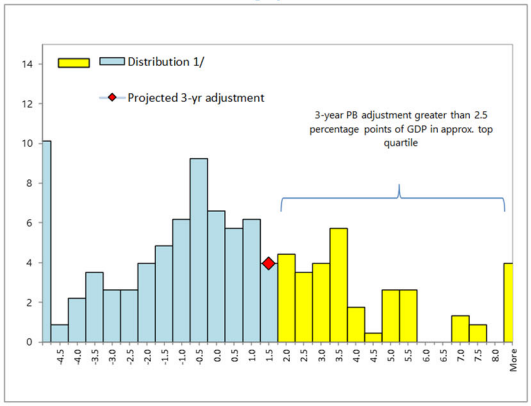
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

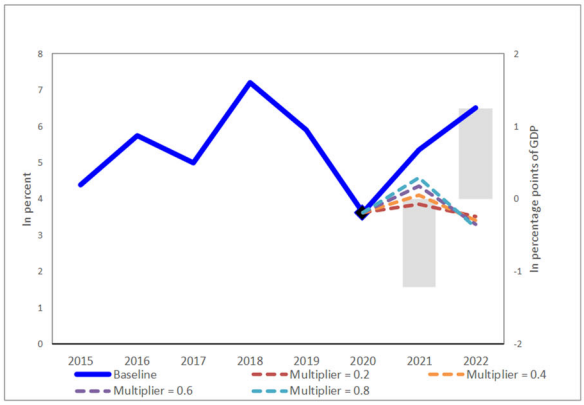
Figure 4. Niger: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



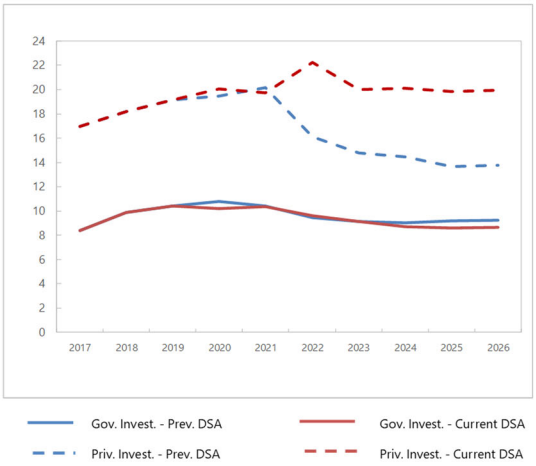
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(% of GDP)



Contribution to Real GDP growth
(percent, 5-year average)

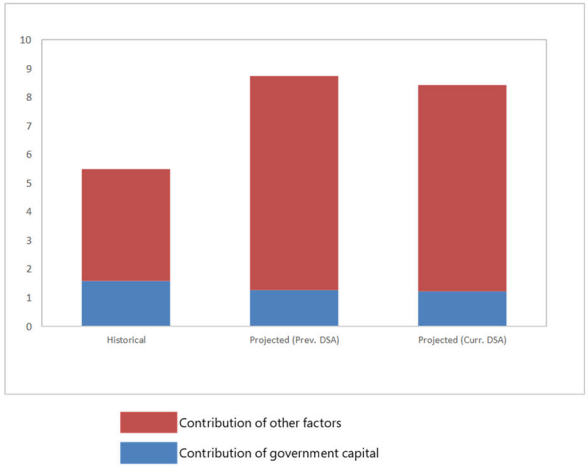
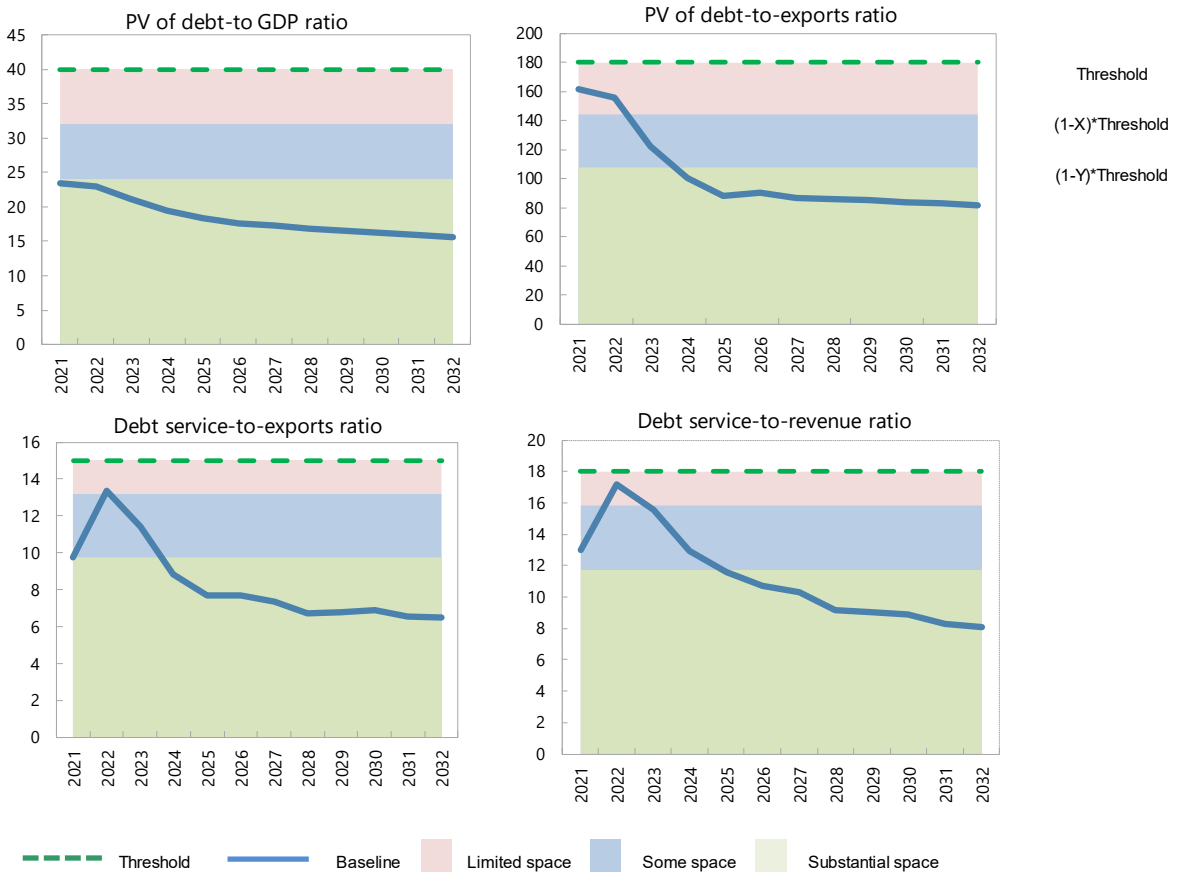


Figure 5. Niger: Qualification of the Moderate Category, 2021–32[1/]



Sources: Country authorities; and staff estimates and projections.

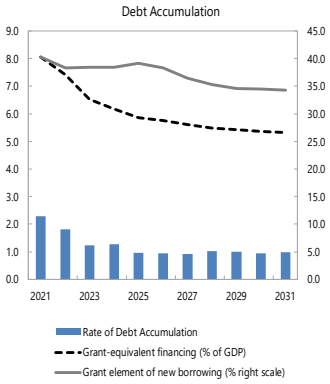
1/ For the PV debt/GDP and PV debt/exports thresholds, x is 20 percent and y is 40 percent. For debt service/Exports and debt service/revenue thresholds, x is 12 percent and y is 35 percent.

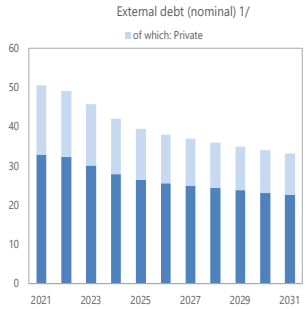
Table 1. Niger: External Debt Sustainability Framework, Baseline Scenario, 2018–41

(In percent of GDP, unless otherwise indicated)

	Actual			Projections												Average 9/	
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2041	Historical	Projections
External debt (nominal) 1/	**45.5**	**47.3**	**50.5**	**50.5**	**49.0**	**45.6**	**42.0**	**39.4**	**38.0**	**36.9**	**35.9**	**34.9**	**34.0**	**33.1**	**24.3**	**43.0**	**39.9**
of which: public and publicly guaranteed (PPG)	**25.3**	**26.5**	**31.6**	**32.8**	**32.2**	**30.0**	**27.9**	**26.3**	**25.5**	**24.9**	**24.3**	**23.7**	**23.1**	**22.6**	**17.9**	**21.9**	**26.7**
Change in external debt	-2.2	1.8	3.2	0.0	-1.5	-3.4	-3.6	-2.6	-1.4	-1.0	-1.0	-1.0	-0.9	-0.9	-1.2		
Identified net debt-creating flows	**3.1**	**7.0**	**7.9**	**8.3**	**8.3**	**3.9**	**1.0**	**1.2**	**2.9**	**2.6**	**2.7**	**2.6**	**2.7**	**2.1**	**2.6**	**5.4**	**3.5**
Non-interest current account deficit	**12.3**	**12.2**	**13.1**	**14.8**	**15.8**	**11.0**	**8.3**	**7.5**	**8.3**	**7.7**	**7.8**	**7.7**	**7.7**	**7.1**	**7.2**	**12.9**	**9.4**
Deficit in balance of goods and services	14.9	16.0	16.2	17.8	18.6	13.6	10.6	9.2	9.5	8.5	8.5	8.2	8.2	7.5	6.9	**14.9**	**10.9**
Exports	11.3	10.8	16.6	14.5	14.8	17.3	19.5	20.8	19.6	20.0	19.6	19.4	19.3	19.1	17.5		
Imports	26.2	26.7	32.8	32.3	33.4	30.9	30.1	30.0	29.0	28.5	28.2	27.5	27.5	26.6	24.5		
Net current transfers (negative = inflow)	-3.8	-4.9	-4.2	-3.9	-3.9	-4.0	-3.9	-3.7	-3.3	-3.2	-3.2	-3.1	-3.0	-3.0	-2.4	**-3.1**	**-3.5**
of which: official	-2.0	-2.8	-2.1	-1.9	-1.7	-1.6	-1.6	-1.6	-1.6	-1.6	-1.6	-1.6	-1.6	-1.6	-1.5		
Other current account flows (negative = net inflow)	1.1	1.1	1.1	1.0	1.1	1.4	1.7	1.9	2.1	2.4	2.5	2.6	2.6	2.6	2.7	**1.1**	**2.0**
Net FDI (negative = inflow)	**-3.3**	**-5.3**	**-2.6**	**-4.6**	**-5.1**	**-3.2**	**-3.3**	**-3.5**	**-3.6**	**-3.4**	**-3.5**	**-3.5**	**-3.5**	**-3.5**	**-3.5**	**-5.6**	**-3.7**
Endogenous debt dynamics 2/	**-5.8**	**0.2**	**-2.6**	**-1.9**	**-2.5**	**-3.9**	**-4.1**	**-2.8**	**-1.8**	**-1.7**	**-1.7**	**-1.6**	**-1.6**	**-1.5**	**-1.2**		
Contribution from nominal interest rate	0.4	0.4	0.4	0.5	0.5	0.5	0.5	0.4	0.4	0.4	0.4	0.3	0.3	0.3	0.3		
Contribution from real GDP growth	-3.0	-2.7	-1.6	-2.4	-3.0	-4.5	-4.5	-3.2	-2.2	-2.1	-2.1	-2.0	-1.9	-1.9	-1.5		
Contribution from price and exchange rate changes	-3.2	2.5	-1.3	...	...	...	...	...	...	...	...	...	...	...	...		
Residual 3/	**-5.3**	**-5.3**	**-4.7**	**-8.4**	**-9.7**	**-7.2**	**-4.6**	**-3.7**	**-4.3**	**-3.6**	**-3.7**	**-3.6**	**-3.6**	**-3.0**	**-3.8**	**-4.0**	**-5.0**
of which: exceptional financing 4/	0.0	0.0	-0.4	-0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators																	
PV of PPG external debt-to-GDP ratio	**...**	**...**	**24.2**	**23.4**	**23.0**	**21.2**	**19.5**	**18.3**	**17.7**	**17.2**	**16.9**	**16.5**	**16.2**	**15.9**	**13.1**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**145.8**	**161.3**	**155.4**	**122.4**	**100.2**	**88.1**	**90.4**	**86.3**	**85.9**	**85.4**	**83.7**	**83.0**	**74.7**		
PPG debt service-to-exports ratio	**7.8**	**7.5**	**6.7**	**9.8**	**13.4**	**11.4**	**8.9**	**7.7**	**7.7**	**7.3**	**6.7**	**6.8**	**6.9**	**6.5**	**5.7**		
PPG debt service-to-revenue ratio	**7.3**	**7.2**	**10.4**	**13.0**	**17.2**	**15.5**	**12.9**	**11.6**	**10.7**	**10.3**	**9.1**	**9.0**	**8.9**	**8.3**	**6.0**		
Gross external financing need (Million of U.S. dollars)	1293.1	1022.6	1646.2	1857.4	2216.9	1955.6	1532.1	1402.3	1702.1	1699.3	1835.9	1925.5	2105.6	1988.8	4396.4		
Key macroeconomic assumptions																	
Real GDP growth (in percent)	7.2	5.9	3.6	5.4	6.5	10.4	11.4	8.5	6.0	5.9	6.1	6.0	6.0	6.0	6.5	**5.7**	**7.1**
GDP deflator in US dollar terms (change in percent)	7.2	-5.1	2.9	7.5	3.2	3.5	3.2	3.1	2.8	2.0	2.0	2.0	2.0	2.0	3.9	**0.3**	**3.0**
Effective interest rate (percent) 5/	0.9	0.8	0.8	1.1	1.2	1.2	1.1	1.1	1.1	1.1	1.0	1.0	1.1	1.1	1.2	**0.8**	**1.1**
Growth of exports of G&S (US dollar terms, in percent)	0.9	-4.2	64.4	-1.0	11.7	33.7	29.6	19.3	2.6	10.2	6.4	6.5	7.9	7.0	7.3	**8.0**	**12.2**
Growth of imports of G&S (US dollar terms, in percent)	14.8	2.4	30.9	11.4	13.5	5.9	11.8	11.5	5.6	6.0	6.9	5.7	7.9	4.6	7.2	**5.7**	**8.2**
Grant element of new public sector borrowing (in percent)	...	...	...	40.3	38.3	38.4	38.4	39.1	38.3	36.4	35.3	34.6	34.4	34.2	32.9	**...**	**37.1**
Government revenues (excluding grants, in percent of GDP)	12.1	11.2	10.8	10.9	11.5	12.7	13.4	13.8	14.0	14.2	14.4	14.7	14.9	15.1	16.5	**11.6**	**13.6**
Aid flows (in Million of US dollars) 6/	1072.2	1399.3	1672.6	1436.8	1427.2	1452.1	1585.7	1683.7	1779.6	1862.7	1972.6	2111.0	2259.0	2421.1	5145.9		
Grant-equivalent financing (in percent of GDP) 7/	...	...	...	8.1	7.4	6.5	6.2	5.8	5.7	5.6	5.5	5.4	5.4	5.3	5.0	**...**	**6.1**
Grant-equivalent financing (in percent of external financing) 7/	...	...	...	77.2	74.4	76.1	76.3	78.6	78.9	78.8	78.8	78.8	78.9	79.0	80.8	**...**	**77.8**
Nominal GDP (Million of US dollars)	12,850	12,912	13,761	15,584	17,122	19,557	22,489	25,163	27,407	29,592	32,007	34,588	37,401	40,455	92,202		
Nominal dollar GDP growth	14.9	0.5	6.6	13.2	9.9	14.2	15.0	11.9	8.9	8.0	8.2	8.1	8.1	8.2	10.6	**6.0**	**10.3**
Memorandum items:																	
PV of external debt 8/	...	...	43.1	41.1	39.8	36.8	33.6	31.3	30.1	29.2	28.5	27.7	27.0	26.4	19.5		
In percent of exports	...	...	259.7	282.8	269.2	212.7	172.5	150.9	153.9	146.5	145.0	143.1	139.9	138.1	111.4		
Total external debt service-to-exports ratio	10.0	9.8	8.7	11.7	15.1	12.7	9.1	7.9	7.9	7.5	6.9	6.9	7.0	6.6	5.7		
PV of PPG external debt (in Million of US dollars)			3333.6	3648.9	3930.4	4139.9	4389.8	4605.5	4844.0	5098.3	5401.1	5719.5	6044.9	6413.1	12060.5		
(PVt-PVt-1)/GDPt-1 (in percent)				2.3	1.8	1.2	1.3	1.0	0.9	0.9	1.0	1.0	0.9	1.0	0.9		
Non-interest current account deficit that stabilizes debt ratio	14.4	10.4	9.9	14.9	17.3	14.4	12.0	10.0	9.7	8.7	8.9	8.7	8.6	8.0	8.5		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	Yes





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ The CCRT debt relief is reflected in the exceptional financing.

5/ Current-year interest payments divided by previous period debt stock.

6/ Defined as grants, concessional loans, and debt relief.

7/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

8/ Assumes that PV of private sector debt is equivalent to its face value.

9/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

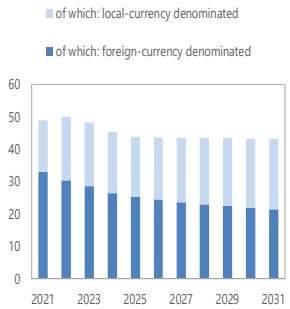
Table 2. Niger: Public Sector Debt Sustainability Framework, Baseline Scenario, 2018–41

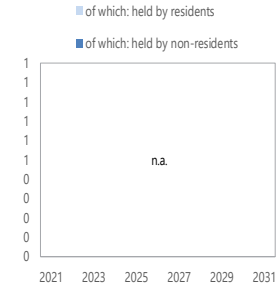
(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 7/	
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2031	2041	Historical	Projections
Public sector debt 1/	**36.9**	**39.8**	**45.0**	**48.8**	**49.8**	**48.2**	**45.3**	**43.8**	**43.5**	**43.1**	**41.3**	**29.5**	**45.1**
of which: external debt	**25.3**	**26.5**	**31.6**	**32.8**	**32.2**	**30.0**	**27.9**	**26.3**	**25.5**	**22.6**	**17.9**	**21.9**	**26.7**
Change in public sector debt	0.4	2.9	5.2	3.8	1.0	-1.6	-2.9	-1.5	-0.3	-0.1	-0.9		
Identified debt-creating flows	**0.7**	**2.0**	**1.3**	**3.8**	**1.0**	**-1.6**	**-3.0**	**-1.5**	**-0.4**	**-0.3**	**-1.0**	**2.2**	**-0.3**
Primary deficit	**2.1**	**2.6**	**4.3**	**5.5**	**4.3**	**3.0**	**1.7**	**1.7**	**1.7**	**1.5**	**1.3**	**3.2**	**2.3**
Revenue incl. grants	18.1	18.0	17.6	17.4	17.3	17.9	18.4	18.7	18.8	19.7	20.9	**16.6**	**18.6**
of which: grants	6.0	6.8	6.8	6.5	5.8	5.2	5.0	4.8	4.8	4.6	4.4		
Primary (noninterest) expenditure	20.2	20.6	21.8	22.9	21.6	20.9	20.1	20.3	20.5	21.2	22.2	**19.9**	**21.0**
Automatic debt dynamics	**-1.4**	**-0.6**	**-3.0**	**-1.7**	**-3.3**	**-4.6**	**-4.7**	**-3.2**	**-2.1**	**-1.7**	**-2.3**		
Contribution from interest rate/growth differential	-2.3	-1.6	-0.8	-2.6	-2.9	-4.3	-4.5	-3.1	-1.9	-1.8	-2.0		
of which: contribution from average real interest rate	0.2	0.5	0.5	-0.3	0.1	0.4	0.4	0.5	0.5	0.7	0.5		
of which: contribution from real GDP growth	-2.5	-2.1	-1.4	-2.3	-3.0	-4.7	-4.9	-3.5	-2.5	-2.4	-2.6		
Contribution from real exchange rate depreciation	0.9	1.0	-2.2	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other) 2/	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**-0.3**	**0.9**	**3.9**	**0.9**	**-0.4**	**-0.3**	**-0.2**	**-0.1**	**0.0**	**0.2**	**-0.1**	**0.8**	**0.1**
Sustainability indicators													
PV of public debt-to-GDP ratio 3/	**...**	**...**	**36.1**	**39.5**	**40.4**	**39.2**	**36.8**	**35.7**	**35.6**	**36.3**	**36.5**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**205.7**	**227.2**	**233.3**	**218.8**	**200.2**	**191.2**	**189.0**	**184.6**	**174.4**		
Debt service-to-revenue and grants ratio 4/	**30.7**	**37.1**	**45.9**	**39.3**	**50.5**	**61.1**	**59.6**	**58.5**	**59.6**	**57.0**	**44.1**		
Gross financing need 5/	6.7	8.3	12.3	12.3	13.0	13.9	12.7	12.6	12.9	12.7	10.5		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	7.2	5.9	3.6	5.4	6.5	10.4	11.4	8.5	6.0	6.0	6.5	**5.7**	**7.1**
Average nominal interest rate on external debt (in percent)	1.6	1.3	1.2	1.5	1.6	1.6	1.5	1.5	1.5	1.5	1.6	**1.5**	**1.5**
Average real interest rate on domestic debt (in percent)	3.5	5.3	4.2	2.6	3.0	3.2	3.5	3.8	3.9	4.2	2.7	**3.8**	**3.7**
Real exchange rate depreciation (in percent, + indicates depreciation)	3.9	4.1	-8.4	...	...	...	...	...	...	...	...	**1.1**	**...**
Inflation rate (GDP deflator, in percent)	2.4	0.1	0.9	2.7	2.0	2.0	2.0	2.0	2.0	2.0	3.9	**1.7**	**2.1**
Growth of real primary spending (deflated by GDP deflator, in percent)	15.0	8.2	10.0	10.3	0.5	6.8	7.2	9.9	6.6	6.5	7.3	**11.0**	**6.8**
Primary deficit that stabilizes the debt-to-GDP ratio 6/	1.6	-0.3	-0.9	1.7	3.3	4.6	4.7	3.2	2.0	1.5	2.1	**0.1**	**2.5**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	Yes

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government plus extra budgetary funds, central bank, government-guaranteed debt. Definition of external debt is Currency-based.

2/ The CCRT debt relief is included in the primary deficit.

3/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

5/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

6/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

7/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 3. Niger: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2021–32

	Projections 1/											
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
PV of debt-to GDP ratio												
Baseline	23	23	21	20	18	18	17	17	17	16	16	16
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2021-2041 2/	23	22	21	22	24	25	27	28	30	31	33	35
B. Bound Tests												
B1. Real GDP growth	23	24	25	23	21	21	20	20	19	19	18	18
B2. Primary balance	23	23	22	20	19	19	18	18	18	17	17	17
B3. Exports	23	26	30	28	26	25	24	23	23	22	21	20
B4. Other flows 3/	23	26	27	25	23	23	22	21	21	20	19	19
B6. One-time 30 percent nominal depreciation	23	29	22	21	19	19	18	18	18	18	18	17
B6. Combination of B1-B5	23	29	29	26	25	24	23	23	22	21	21	20
C. Tailored Tests												
C1. Combined contingent liabilities	23	24	23	21	20	20	19	19	19	18	18	18
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	23	24	24	22	21	20	19	18	17	17	16	15
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	40	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio												
Baseline	161	155	122	100	88	90	86	86	85	84	83	82
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2021-2041 2/	161	146	121	114	115	128	133	144	154	162	174	183
B. Bound Tests												
B1. Real GDP growth	161	155	122	100	88	90	86	86	85	84	83	82
B2. Primary balance	161	157	126	104	92	95	91	91	91	89	89	87
B3. Exports	161	225	350	285	250	256	243	240	235	226	220	213
B4. Other flows 3/	161	177	158	129	113	116	110	109	107	103	100	98
B6. One-time 30 percent nominal depreciation	161	155	103	84	74	76	73	73	74	73	73	73
B6. Combination of B1-B5	161	207	141	184	162	166	158	156	154	150	147	144
C. Tailored Tests												
C1. Combined contingent liabilities	161	162	130	107	96	100	96	96	96	95	94	93
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	161	186	150	120	103	103	96	94	91	87	85	82
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	180	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio												
Baseline	10	13	11	9	8	8	7	7	7	7	7	6
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2021-2041 2/	10	14	12	10	10	10	10	9	9	10	11	11
B. Bound Tests												
B1. Real GDP growth	10	13	11	9	8	8	7	7	7	7	7	6
B2. Primary balance	10	13	11	9	8	8	7	7	7	7	7	7
B3. Exports	10	18	24	20	18	18	17	16	19	19	18	17
B4. Other flows 3/	10	13	12	10	8	8	8	8	9	9	8	8
B6. One-time 30 percent nominal depreciation	10	13	11	8	7	7	7	6	6	6	6	6
B6. Combination of B1-B5	10	15	19	15	13	13	12	12	12	12	12	12
C. Tailored Tests												
C1. Combined contingent liabilities	10	13	12	9	8	8	8	7	7	7	7	7
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	10	15	13	10	9	8	8	8	8	8	7	7
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	15	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio												
Baseline	13	17	16	13	12	11	10	9	9	9	8	8
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2021-2041 2/	13	18	17	15	15	14	14	12	12	13	13	14
B. Bound Tests												
B1. Real GDP growth	13	18	18	15	13	12	12	11	10	10	10	9
B2. Primary balance	13	17	16	13	12	11	10	9	9	9	9	8
B3. Exports	13	17	16	15	13	12	12	11	12	12	11	11
B4. Other flows 3/	13	17	16	14	12	12	11	11	11	11	10	10
B6. One-time 30 percent nominal depreciation	13	22	20	16	14	13	12	11	10	10	9	9
B6. Combination of B1-B5	13	19	19	16	14	13	13	12	12	12	11	11
C. Tailored Tests												
C1. Combined contingent liabilities	13	17	16	13	12	11	11	9	9	9	9	8
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	13	21	19	16	14	12	11	10	10	10	9	9
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	18	18	18	18	18	18	18	18	18	18	18	18

Sources: Country authorities; and staff estimates and projections.

1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Niger: Sensitivity Analysis for Key Indicators of Public Debt, 2021–32

	Projections											
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
PV of Debt-to-GDP Ratio												
Baseline	**39**	**40**	**39**	**37**	**36**	**36**	**36**	**36**	**36**	**36**	**36**	**36**
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2018-2038 1/	39	40	41	41	42	43	44	45	47	48	49	50
B. Bound Tests												
B1. Real GDP growth	39	43	48	47	47	49	51	53	55	56	58	60
B2. Primary balance	39	42	42	40	38	38	38	38	38	38	38	38
B3. Exports	39	43	47	44	42	42	42	42	41	41	41	40
B4. Other flows 2/	39	44	45	42	41	41	40	40	40	40	40	39
B6. One-time 30 percent nominal depreciation	39	45	42	38	35	34	33	32	31	31	30	29
B6. Combination of B1-B5	39	40	41	39	38	38	38	38	39	39	39	39
C. Tailored Tests												
C1. Combined contingent liabilities	39	47	45	42	40	40	40	40	40	40	40	39
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	39	43	46	48	51	54	57	58	60	62	64	66
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Public debt benchmark	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**
PV of Debt-to-Revenue Ratio												
Baseline	227	233	219	200	191	189	189	187	187	185	185	184
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2018-2038 1/	227	229	223	217	217	219	224	227	231	234	238	242
B. Bound Tests												
B1. Real GDP growth	227	244	256	245	244	250	258	265	272	277	284	290
B2. Primary balance	227	243	236	215	205	201	200	198	196	194	193	191
B3. Exports	227	247	263	239	227	223	221	218	214	210	207	204
B4. Other flows 2/	227	251	253	230	219	215	214	211	208	204	202	199
B6. One-time 30 percent nominal depreciation	227	269	240	211	194	185	178	172	166	160	155	151
B6. Combination of B1-B5	227	235	227	209	201	199	200	199	199	198	198	198
C. Tailored Tests												
C1. Combined contingent liabilities	227	271	251	228	216	212	210	207	205	202	201	199
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	227	276	286	289	290	297	302	302	309	315	322	328
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio												
Baseline	39	51	61	60	58	60	62	57	57	57	57	59
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2018-2038 1/	39	51	61	64	63	66	69	64	65	66	66	69
B. Bound Tests												
B1. Real GDP growth	39	52	69	72	74	80	86	83	87	88	91	95
B2. Primary balance	39	51	66	66	64	66	67	61	61	60	60	62
B3. Exports	39	51	61	61	59	60	62	58	60	59	59	61
B4. Other flows 2/	39	51	62	60	59	60	62	58	59	59	59	61
B6. One-time 30 percent nominal depreciation	39	50	62	60	58	59	61	56	56	56	55	57
B6. Combination of B1-B5	39	50	62	63	62	64	67	62	63	63	63	66
C. Tailored Tests												
C1. Combined contingent liabilities	39	51	79	68	72	70	69	64	64	62	62	63
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	39	57	73	81	87	94	101	97	100	101	103	107
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.

2/ Includes official and private transfers and FDI.